SMART CONTRACT AUDIT REPORT

for

ArcNova Token

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1 Introduction

Given the opportunity to review the design document and related source code of the ArcNova token contract, we outline in the report our systematic method to evaluate potential security issues in the smart contract implementation, expose possible semantic inconsistency between smart contract code and the documentation, and provide additional suggestions or recommendations for improvement. Our results show that the given version of the smart contract exhibits no ERC20 compliance issues or security concerns. This document outlines our audit results.

1.1 About ArcNova Token

ArcNova is a non-upgradeable ERC20 token contract with a fixed supply of 1,000,000,000. It has been designed with no mint, no burn, no pause, no blacklist, and no proxy/upgrade mechanism. It also has a companion config-only wrapper contract around the official LayerZero V2 OFTAdapter that is used to lock ArcNova tokens on Ethereum when sending to BSC and unlock on the way back. This audit examines the token contract implementation with a focus on the compatibility of the ERC20 specification as well as other known ERC20 pitfalls and vulnerabilities. The basic information of the audited contract is as follows:

Table 1.1: Basic Information of ArcNova Token Contract

Item	Description
Name	ArcNova
Type	ERC20 Token Contract
Platform	Solidity
Audit Method	Whitebox
Audit Completion Date	July 15, 2026

In the following, we show the Git repository of reviewed files and the commit hash value used in this audit. Note this repository also contains ArcNovaOFTAdapter and BridgedAN contracts that are thin constructor wrappers around LayerZero V2's official OFTAdapter / OFT abstract contracts. Further, it

has a companion multi-beneficiary linear vesting manager contract. These contracts are also covered in this audit.

- https://github.com/ArcNova-ACI/arcnova-token.git (42cecf5)

1.2 About PeckShield

PeckShield Inc. [4] is a leading blockchain security company with the goal of elevating the security, privacy, and usability of current blockchain ecosystem by offering top-notch, industry-leading services and products (including the service of smart contract auditing). We are reachable at Telegram (https://t.me/peckshield), Twitter (http://twitter.com/peckshield), or Email (contact@peckshield.com).

1.3 Methodology

To standardize the evaluation, we define the following terminology based on OWASP Risk Rating Methodology [3]:

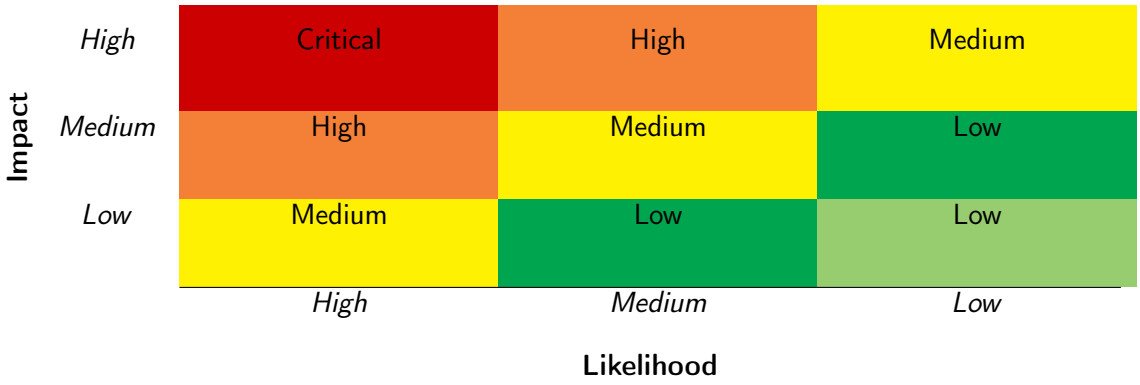
- Likelihood represents how likely a particular vulnerability is to be uncovered and exploited in the wild;
- Impact measures the technical loss and business damage of a successful attack;
- Severity demonstrates the overall criticality of the risk;

Likelihood and impact are categorized into three ratings: *H*, *M* and *L*, i.e., *high*, *medium* and *low* respectively. Severity is determined by likelihood and impact and can be classified into four categories accordingly, i.e., *Critical*, *High*, *Medium*, *Low* shown in Table 1.2.

We perform the audit according to the following procedures:

- Basic Coding Bugs: We first statically analyze given smart contracts with our proprietary static code analyzer for known coding bugs, and then manually verify (reject or confirm) all the issues found by our tool.
- ERC20 Compliance Checks: We then manually check whether the implementation logic of the audited smart contract(s) follows the standard ERC20 specification and other best practices.
- Additional Recommendations: We also provide additional suggestions regarding the coding and development of smart contracts from the perspective of proven programming practices.

Table 1.2: Vulnerability Severity Classification



To evaluate the risk, we go through a list of check items and each would be labeled with a severity category. For one check item, if our tool does not identify any issue, the contract is considered safe regarding the check item. For any discovered issue, we might further deploy contracts on our private testnet and run tests to confirm the findings. If necessary, we would additionally build a PoC to demonstrate the possibility of exploitation. The concrete list of check items is shown in Table 1.3.

1.4 Disclaimer

Note that this security audit is not designed to replace functional tests required before any software release, and does not give any warranties on finding all possible security issues of the given smart contract(s) or blockchain software, i.e., the evaluation result does not guarantee the nonexistence of any further findings of security issues. As one audit-based assessment cannot be considered comprehensive, we always recommend proceeding with several independent audits and a public bug bounty program to ensure the security of smart contract(s). Last but not least, this security audit should not be used as investment advice.

Table 1.3: The Full List of Check Items

Category	Check Item
Basic Coding Bugs	Constructor Mismatch
	Ownership Takeover
	Redundant Fallback Function
	Overflows & Underflows
	Reentrancy
	Money-Giving Bug
	Blackhole
	Unauthorized Self-Destruct
	Revert DoS
	Unchecked External Call
	Gasless Send
	Send Instead of Transfer
	Costly Loop
	(Unsafe) Use of Untrusted Libraries
	(Unsafe) Use of Predictable Variables
	Transaction Ordering Dependence
	Deprecated Uses
	Approve / TransferFrom Race Condition
ERC20 Compliance Checks	Compliance Checks (Section 3)
Additional Recommendations	Avoiding Use of Variadic Byte Array
	Using Fixed Compiler Version
	Making Visibility Level Explicit
	Making Type Inference Explicit
	Adhering To Function Declaration Strictly
	Following Other Best Practices

2 | Findings

2.1 Summary

Here is a summary of our findings after analyzing the `ArcNova` token contract. During the first phase of our audit, we study the smart contract source code and run our in-house static code analyzer through the codebase. The purpose here is to statically identify known coding bugs, and then manually verify (reject or confirm) issues reported by our tool. We further manually review business logics, examine system operations, and place ERC20-related aspects under scrutiny to uncover possible pitfalls and/or bugs.

Severity	# of Findings	
Critical	0	
High	0	
Medium	0	
Low	0	
Informational	1	■
Total	1	

Moreover, we explicitly evaluate whether the given contracts follow the standard ERC20 specification and other known best practices, and validate its compatibility with other similar ERC20 tokens and current DeFi protocols. The detailed ERC20 compliance checks are reported in Section 3. After that, we examine a few identified issues of varying severities that need to be brought up and paid more attention to. (The findings are categorized in the above table.) Additional information can be found in the next subsection, and the detailed discussions are in Section 4.

2.2 Key Findings

Overall, no ERC20 compliance issue was found and our detailed checklist can be found in Section 3. While there is no critical or high severity issue, the implementation can be improved by resolving the identified issues, including 1 informational recommendation (shown in Table 2.1).

Table 2.1: Key ArcNova Token Audit Findings

ID	Severity	Title	Category	Status
PVE-001	Informational	Strengthened Schedule Duration Validation in VestingManager	Coding Practices	Resolved

Besides recommending specific countermeasures to mitigate the above issue(s), we also emphasize that it is always important to develop necessary risk-control mechanisms and make contingency plans, which may need to be exercised before the mainnet deployment. The risk-control mechanisms need to kick in at the very moment when the contracts are being deployed in mainnet. Please refer to Section 3 for our detailed compliance checks and Section 4 for elaboration of reported issues.

3 | ERC20 Compliance Checks

The ERC20 specification defines a list of API functions (and relevant events) that each token contract is expected to implement (and emit). The failure to meet these requirements means the token contract cannot be considered to be ERC20-compliant. Naturally, as the first step of our audit, we examine the list of API functions defined by the ERC20 specification and validate whether there exist any inconsistency or incompatibility in the implementation or the inherent business logic of the audited contract(s).

Table 3.1: Basic View-Only Functions Defined in The ERC20 Specification

Item	Description	Status
name()	Is declared as a public view function	✓
	Returns a string, for example "Tether USD"	✓
symbol()	Is declared as a public view function	✓
	Returns the symbol by which the token contract should be known, for example "USDT". It is usually 3 or 4 characters in length	✓
decimals()	Is declared as a public view function	✓
	Returns decimals, which refers to how divisible a token can be, from 0 (not at all divisible) to 18 (pretty much continuous) and even higher if required	✓
totalSupply()	Is declared as a public view function	✓
	Returns the number of total supplied tokens, including the total minted tokens (minus the total burned tokens) ever since the deployment	✓
balanceOf()	Is declared as a public view function	✓
	Anyone can query any address' balance, as all data on the blockchain is public	✓
allowance()	Is declared as a public view function	✓
	Returns the amount which the spender is still allowed to withdraw from the owner	✓

Our analysis shows that there is no ERC20 inconsistency or incompatibility issue found in the audited token contract. In the surrounding two tables, we outline the respective list of basic view-only functions (Table 3.1) and key state-changing functions (Table 3.2) according to the widely-adopted

ERC20 specification.

Table 3.2: Key State-Changing Functions Defined in The ERC20 Specification

Item	Description	Status
transfer()	Is declared as a public function	✓
	Returns a boolean value which accurately reflects the token transfer status	✓
	Reverts if the caller does not have enough tokens to spend	✓
	Allows zero amount transfers	✓
	Emits Transfer() event when tokens are transferred successfully (include 0 amount transfers)	✓
transferFrom()	Is declared as a public function	✓
	Returns a boolean value which accurately reflects the token transfer status	✓
	Reverts if the spender does not have enough token allowances to spend	✓
	Updates the spender's token allowances when tokens are transferred successfully	✓
	Reverts if the from address does not have enough tokens to spend	✓
	Allows zero amount transfers	✓
	Emits Transfer() event when tokens are transferred successfully (include 0 amount transfers)	✓
approve()	Is declared as a public function	✓
	Returns a boolean value which accurately reflects the token approval status	✓
	Emits Approval() event when tokens are approved successfully	✓
Transfer() event	Is emitted when tokens are transferred, including zero value transfers	✓
	Is emitted with the from address set to $address(0x0)$ when new tokens are generated	✓
Approve() event	Is emitted on any successful call to approve()	✓

In addition, we perform a further examination on certain features that are permitted by the ERC20 specification or even further extended in follow-up refinements and enhancements (e.g., ERC777), but not required for implementation. These features are generally helpful, but may also impact or bring certain incompatibility with current DeFi protocols. Therefore, we consider it is important to highlight them as well. This list is shown in Table 3.3.

Meanwhile, the associated BridgedAN token contract is implemented as an official LayerZero V2 OFT, where the supply on BSC is minted/burned exclusively by verified LayerZero messages coming from the Ethereum OFTAdapter and there is no public or owner mint. Total supply here always mirrors the amount locked in the Ethereum adapter.

Table 3.3: Additional Opt-in Features Examined in Our Audit

Feature	Description	Opt-in
Deflationary	Part of the tokens are burned or transferred as a fee while on transfer()/transferFrom() calls	—
Rebasing	The balanceOf() function returns a re-based balance instead of the actual stored amount of tokens owned by the specific address	—
Pausable	The token contract allows the owner or privileged users to pause the token transfers and other operations	—
Blacklistable	The token contract allows the owner or privileged users to blacklist a specific address such that token transfers and other operations related to that address are prohibited	—
Mintable	The token contract allows the owner or privileged users to mint tokens to a specific address	—
Burnable	The token contract allows the owner or privileged users to burn tokens of a specific address	—

4 Detailed Results

4.1 Strengthened Schedule Duration Validation in VestingManager

- ID: PVE-001
- Severity: Informational
- Likelihood: N/A
- Impact: N/A
- Target: `VestingManager`
- Category: Security Features [2]
- CWE subcategory: CWE-287 [1]

Description

The `ArcNova` token contract has a companion multi-beneficiary linear vesting manager contract named `VestingManager`. In the process of reviewing its logic to create vesting schedules for a given beneficiary, we notice the given vesting schedule duration may be more thoroughly validated.

To elaborate, we show below the implementation of the related function, i.e., `createSchedule()`. It has a rather straightforward logic in validating the given parameter and then creating an immutable vesting schedule. We notice the given vesting ending time should be reliably calculated from `start + duration` without being overflowed. In other words, we can improve current validation by adding the following statement, i.e., `if (start > type(uint256).max - duration)revert IncorrectSchedule();`.

```
84      function createSchedule(
85          bytes32 scheduleId,
86          address beneficiary,
87          uint256 totalAmount,
88          uint256 start,
89          uint256 cliff,
90          uint256 duration
91      ) external onlyOwner {
92          if (beneficiary == address(0)) revert BeneficiaryIsZeroAddress();
93          if (totalAmount == 0) revert TotalAmountIsZero();
94          if (duration == 0) revert DurationIsZero();
95          if (cliff > duration) revert CliffExceedsDuration();
96          if (start > type(uint256).max - duration) revert IncorrectScheduleDuration();
```

```
97          if (schedules[scheduleId].exists) revert ScheduleAlreadyExists(scheduleId);
98
99          uint256 unallocated = token.balanceOf(address(this)) + totalReleased -
                totalAllocated;
100         if (unallocated < totalAmount) revert InsufficientUnallocatedBalance(unallocated
                , totalAmount);
101
102         schedules[scheduleId] = VestingSchedule({
103             beneficiary: beneficiary,
104             totalAmount: totalAmount,
105             start: start,
106             cliff: cliff,
107             duration: duration,
108             released: 0,
109             exists: true
110         });
111         totalAllocated += totalAmount;
112
113         emit ScheduleCreated(scheduleId, beneficiary, totalAmount, start, cliff,
                duration);
114     }
```

Listing 4.1: `VestingManager::createSchedule()`

Recommendation Revise the above routine to ensure the given parameters for a vesting schedule is sound.

Status This issue has been resolved. The team clarifies that the only caller to create a vesting schedule is the trusted `owner`.

5 | Conclusion

In this security audit, we have examined the ArcNova token contract design and implementation. During our audit, we first checked all respects related to the compatibility of the ERC20 specification and other known ERC20 pitfalls/vulnerabilities and found no issue in these areas. We then proceeded to examine other areas such as coding practices and business logics. Overall, no issue was found in these areas, and the current deployment follows the best practice. Meanwhile, as disclaimed in Section 1.4, we appreciate any constructive feedbacks or suggestions about our findings, procedures, audit scope, etc.

References

[1] MITRE. CWE-287: Improper Authentication. https://cwe.mitre.org/data/definitions/287.html.

[2] MITRE. CWE CATEGORY: 7PK - Security Features. https://cwe.mitre.org/data/definitions/254.html.

[3] OWASP. Risk Rating Methodology. https://www.owasp.org/index.php/OWASP_Risk_Rating_Methodology.

[4] PeckShield. PeckShield Inc. https://www.peckshield.com.